The Identifiability Gap in Autonomous Research: When Can Systems Be Ranked?

Hongmin Li, Institute of Science Tokyo; The University of Tokyo

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Abstract

Autonomous-research systems are usually compared through selected papers, proofs, and discoveries. We ask when such observations identify per-attempt reliability and the order of systems. We define reliability as $\theta_{\mathcal{D},B,\pi} = P(V = 1 \mid \text{attempt initiated under } \mathcal{D}, B, \pi)$, keeping failed and no-output attempts in the denominator. Adapting classical partial-identification arguments, we show that a selected success rate identifies neither θ nor a system order, give sharp missing-outcome bounds for bounded endpoints, state an order-identification criterion for separable identified sets, and give sufficient design conditions for point identification. We then audit public evidence. A screen of 45 papers (a 44-paper evidence map plus one recent benchmark) yields 15 endpoint-specific comparative evaluations with 100 configurations and 376 within-evaluation pairs. In the four evaluations whose displays rest on selected or partially observed outputs, 119 of 134 displayed strict orders survive denominator recovery, six reverse direction, and nine become unresolved; in MLReplicate, none of the 15 pairwise system orders is identified. The 11 complete-run or near-complete evaluations retain all 214 displayed strict orders, as they must when display and target coincide. "Certified" means retained for the realized endpoint, which may be a proxy for scientific validity; a certified order can still be small or statistically uncertain. Reversals show that best-of-k orders need not transfer to the all-attempt target, not that the source's own results are wrong. These are finite-design results: pairs are dependent, the evidence map is purposive, the audit was not preregistered, and no field prevalence or global "illusory-capability fraction" is estimated. The practical consequence is a rankability certificate that reports only the partial order supported by the observation design. All data, reconstruction code, and independent numerical checks are in the supplement.

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Problem and contribution

Question. An autonomous-research system is run on tasks; some attempts fail, some outputs are selected for display or validation, and a possibly fallible validator labels what is shown. When do these public observations identify how reliable the system is per initiated attempt, and when do they identify which of two systems is more reliable?

Established without further assumptions (Section 3). (i) A selected validity rate is compatible with every per-attempt reliability in $(0, 1)$ (Proposition 1). (ii) Given completion, selection, and selected-mean observations, the sharp identified set for a bounded endpoint is an explicit interval, narrowing under monotone selection (Proposition 2). (iii) Under separable restrictions, a strict order between two systems is identified exactly when the lower endpoint of one set exceeds the upper endpoint of the other (Proposition 3). These are elementary adaptations of known partial-identification results [1, 2]; we do not claim new statistical theory.

Established on the released data (Section 5). The 15-evaluation rankability audit and the MLReplicate reconstruction are deterministic functions of source-reported counts and scores; the supplement recomputes every reported count.

Conditional. Validator correction (Section 3.5) requires known and transportable sensitivity and specificity. Point identification of $\theta_{\mathcal{D},B,\pi}$ (Proposition 4) requires a complete denominator, a common reference procedure, and either probability sampling from $\mathcal{D}$ or an explicit transport model.

Not claimed. No estimate of field prevalence, no pooled capability effect, no global fraction of capability that is a post-selection artifact, no claim that leaderboards generally fail, and no ranking of underlying architectures at fixed compute. The coding-sensitivity and validator-separation analyses of Section 5.5 are descriptive and partly exploratory.

1. Introduction

Systems described as AI scientists now span idea generation, literature search, experimentation, manuscript production, architecture discovery, and research-level mathematics [3, 4, 5, 6]. A proof accepted by specialists or a correctly executed experiment establishes that a system can sometimes contribute to research. It does not establish how reliably that contribution occurs over a target problem distribution, a fixed compute and human-validation budget, and a stated selection policy.

The distinction matters when generation is cheap and validation is scarce. A system can generate many candidates, inspect downstream results, discard unsuccessful runs, and display only favorable artifacts. The visible artifact then conflates at least four quantities: completion, candidate selection, scientific validity, and validator error. A transparent best-of-k procedure may legitimately improve the displayed result; a selector that sees information intended to remain isolated may introduce bias. Pipeline attrition, wrong mechanisms, unsupported claims, and false-negative validators are distinct failure routes, and averaging them into a single "illusory capability fraction" yields a number without a coherent unit or counterfactual.

We study the identification problem behind autonomous-research evaluation. The target is not the validity of an already selected paper but the probability that an initiated attempt produces an artifact satisfying a prespecified validity criterion under task distribution $\mathcal{D}$, budget B, and policy π. This separates what a system can do once, what selection among attempts contributes, and what released observations imply about a target population. It does not attribute control of generation, decision making, or verification inside a successful artifact; those are actor–stage relations rather than attempt-level reliability.

Across 15 reconstructable comparative evaluations, denominator recovery preserves, reverses, or erases displayed relations according to the observation design. Six relations reverse direction rather than merely becoming uncertain. Complete-run evaluations behave as positive controls, and MLReplicate is the limiting empty-graph case [7]. The empirical finding is therefore not that leaderboards generally fail: rankability is a property of what an evaluation observes, not only of the scores it displays.

2. Related work

Systems and evaluations. End-to-end AI-scientist systems and benchmarks cover ML experimentation, architecture search, open-ended trajectories, reproduction, and research mathematics [3, 8, 4, 9, 5, 6]. Independent and benchmark evaluations report execution failures, integrity violations, similarity to prior work, invalid mechanisms, evidence non-uptake, and implementation–claim disconnects [10, 11, 12, 13, 14]. We treat these as different targets rather than interchangeable evidence of post-selection.

Verification. Cornelio et al. argue that scalable hypothesis generation can overwhelm empirical, logical, and human verification, and that benchmarks often fail to test novelty, generalizability, and scientific consistency [15]. Ding et al. distinguish mechanical loops from outside-oracle validation and note that a real experimental result need not validate an attached mechanistic explanation [16]. Validator separation and endpoint reach are therefore not new concepts here; we use them as conditions inside an identification argument and test their consequence for rankability. A previous conceptual paper by the present author frames AI-assisted research as a loop whose output claims must be licensed by evidence [17]; it contains no estimand, no identification result, and no rankability audit, and the present paper does not depend on it.

Partial identification and ranking. Partial identification replaces unsupported point estimates with sets implied by data and explicit restrictions [1, 18]. It can induce only a partial ranking: in policy choice, orderability depends on the identifying restrictions [19], and Crippa and Fedchenko characterize partially identified rankings when merits are inferred from incompletely observed pairwise interactions [20]. Our observation model differs: systems need not interact pairwise, and the missing quantities are attempt-level outcomes hidden by completion, selection, and validation. Sharp bounds under selective attrition [2] and evaluation under selective labels [21] address two of these layers; the third, an uncalibrated validator, is the classical sensitivity–specificity correction [22].

Leaderboard evidence. Oh shows across ten public benchmarks that mean-score winners often lack meaningful effect size, cross-task consistency, or robustness to dataset removal [23]. Leaderboard fragility and calls for reporting discipline are therefore not new. The certificate proposed here is narrower: it carries the autonomous-research estimand and observation chain, separates target incompatibility from missing-attempt ambiguity, and reports only orders that survive denominator recovery.

Relation to rStation papers. At the time of writing the AI agents board contains one paper, a census of agent-card conformance; it concerns protocol measurement, not research-capability evaluation, and does not overlap with this work.

3. Identification framework

3.1 Target estimand and observation process

An attempt draws a task from $\mathcal{D}$, runs policy π under budget B, and targets a prespecified artifact-level validity criterion. Define

$$\theta_{\mathcal{D},B,\pi} = P(V = 1 \mid \text{attempt initiated under } \mathcal{D}, B, \pi),$$

where $V = 1$ means that the resulting artifact satisfies the stated criterion under a sufficiently independent reference procedure. For this qualifying-artifact estimand, no-output and failed execution have $V = 0$. The convention is endpoint-specific: a failed calculation can still produce useful critique, so it must not be used to estimate a broader quantity such as "any scientific value".

The observable chain is $A \to C \to S \to V \to \widehat{V} \to R$: initiation, candidate completion, selection for display or validation, reference validity, a possibly fallible validator label, and public reporting. Most reports observe a conditional quantity such as

$$q_{\text{obs}} = P(\widehat{V} = 1 \mid C = 1, S = 1, R = 1),$$

not $\theta_{\mathcal{D},B,\pi}$. Even complete observation of a curated evaluation set point-identifies only its finite-design reliability; transport to $\mathcal{D}$ requires a declared sampling frame or explicit external-validity assumptions.

3.2 Selected success does not identify reliability

Let S denote selection, $\theta = P(V = 1)$, and $q = P(V = 1 \mid S = 1)$.

Proposition 1 (non-identification from selected success). For any $q, \theta \in (0, 1)$ there is a selection process with $P(S = 1) > 0$, selected validity q, and per-attempt validity θ.

Proof. Let $a = P(S = 1 \mid V = 1)$ and $b = P(S = 1 \mid V = 0)$. If $\theta \le q$, set $a = 1$ and $b = \theta(1 - q)/[q(1 - \theta)]$; then $b \le 1$ because $\theta(1-q) \le q(1-\theta)$. If $\theta > q$, set $b = 1$ and $a = q(1-\theta)/[(1-q)\theta] < 1$. In both cases $P(S = 1) = \theta a+(1-\theta)b > 0$ and

$$P(V = 1 \mid S = 1) = \frac{\theta a}{\theta a + (1 - \theta)b} = q. \qquad \square$$

The interior restriction excludes only boundary incompatibilities: with $P(S = 1) > 0$, $\theta = 0$ forces $q = 0$ and $\theta = 1$ forces $q = 1$. Selected validity alone therefore identifies neither per-attempt reliability nor any global fraction of capability attributable to selection.

3.3 Sharp identified sets

Let $Y \in [0, 1]$ be a bounded endpoint, with $Y = V$ as the binary case. Define completion $c = P(C = 1)$, selection $s = P(S = 1 \mid C = 1)$, selected mean $\mu_s = E[Y \mid C = 1, S = 1]$, and unselected mean $\mu_u = E[Y \mid C = 1, S = 0]$, where $c > 0$ and $s > 0$ so that μ_s is defined. If non-completion has $Y = 0$, then

$$\theta = E[Y] = c\{s\mu_s + (1 - s)\mu_u\}.$$

Proposition 2 (sharp missing-outcome bounds). If c, s, μ_s are identified and $\mu_u \in [0, 1]$ is otherwise unrestricted, the sharp identified set is

$$\theta \in [cs\mu_s,\ c\{s\mu_s + (1 - s)\}].$$

Under the additional monotone-selection restriction $\mu_u \le \mu_s$, the sharp set is $[cs\mu_s,\ c\mu_s]$.

Proof. The decomposition is affine in μ_u, so its range over $[0,1]$ (respectively $[0,\mu_s]$) is attained at the endpoints. Every value in between is attained by assigning the unselected units a common outcome distribution with that mean, which is consistent with all observed quantities. □

These sets describe identification, not sampling uncertainty; confidence regions for partially identified parameters must also cover uncertainty in the estimated endpoints [18].

3.4 Order identification and ranking reversal

Let systems i and j target the same $(\mathcal{D}, B, V)$, with sharp identified sets $I_i = [L_i, U_i]$ and $I_j = [L_j, U_j]$. Call the restrictions *separable* if every pair in $I_i \times I_j$ is jointly attainable, that is, no admissible completion for one system restricts the other.

Proposition 3 (order identification). Under separable restrictions, $\theta_i > \theta_j$ is identified if and only if $L_i > U_j$. If $L_i \le U_j$ and $L_j \le U_i$, neither strict order is identified: when $L_i < U_j$ the values $(\theta_i, \theta_j) = (L_i, U_j)$ give a compatible reversal, and when $L_i = U_j$ they give a compatible tie.

Proof. If $L_i > U_j$, every jointly attainable pair satisfies $\theta_i \ge L_i > U_j \ge \theta_j$. Conversely, if $L_i \le U_j$, separability makes (L_i, U_j) attainable, and it violates $\theta_i > \theta_j$. □

For example, a system with selected validity 0.9, selection rate 0.1, and full completion has $\theta \in [0.09, 0.99]$. A second system evaluated on every attempt with validity 0.6 has $\theta = 0.6$. The first system displays the higher score, yet neither order is identified. We define the *rankability graph* to have a directed edge $i \to j$ only when the systems are target-compatible and $L_i > U_j$; it is a strict partial order, not a forced leaderboard.

For $m \ge 2$ configurations, let E_D contain the strict directed edges implied by the displayed scores and E_I those implied by the denominator-aware identified sets. We summarize strict orderability by

$$\rho = \frac{|E_I|}{\binom{m}{2}},$$

and, when $|E_D| > 0$, the loss of displayed strict claims by

$$G_R = 1 - \frac{|E_D \cap E_I|}{|E_D|},$$

where the intersection requires the same direction. A reversal and an unresolved displayed edge both contribute to G_R; a displayed tie, which makes no strict claim, does not. These are finite-design graph summaries, not confidence measures.

3.5 Validator error and point identification

A complete denominator is insufficient when V is replaced by a noisy label. Let sensitivity $\alpha = P(\widehat{V} = 1 \mid V = 1)$ and specificity $\beta = P(\widehat{V} = 0 \mid V = 0)$, assumed equal for selected and unselected artifacts. Among selected units, with $v_s = P(V = 1 \mid C = 1, S = 1)$,

$$q = P(\widehat{V} = 1 \mid C = 1, S = 1) = \alpha v_s + (1 - \beta)(1 - v_s).$$

If $\alpha + \beta > 1$ and both are known and transportable to the target artifacts, then

$$v_s = \frac{q + \beta - 1}{\alpha + \beta - 1},$$

which lies in $[0,1]$ exactly when $1 - \beta \le q \le \alpha$ [22]. Intervals for α and β induce bounds on v_s. Sensitivity alone does not identify the correction: SciCoQA's best tested validator detects 43 of 92 ascertained real paper–code discrepancies, which identifies a positive-case sensitivity in that benchmark, not a specificity or a correction for AI-generated research [24].

Proposition 4 (a sufficient point-identification design). The realized finite-design mean is point identified if every initiated unit under a fixed policy and budget is retained, the endpoint is observed for all units under one independent reference procedure, and no-output and failed runs receive the prespecified endpoint value. If only $\widehat{V}$ is observed, the validator error rates must also be identified and applicable. Identification of $\theta_{\mathcal{D},B,\pi}$ further requires probability sampling from $\mathcal{D}$ or an explicit transport model.

This is one transparent sufficient design, not the only route. A prespecified probability sample of units with known positive inclusion probabilities supports design-based estimation, with its sampling uncertainty reported. Substantive restrictions can identify other models, but they must be stated and tested rather than embedded in a selected score.

4. Evidence and audit design

4.1 Evidence map

The search was current to 26 August 2026. Nine fixed arXiv API queries over 1 January 2023 to 26 August 2026 returned 446 unique records with complete pagination (100 results per page, sorted by submission date, run on 26 August 2026, each query conjoined with the date filter `submittedDate:[202301010000 TO 202608262359]`). The query strings, with `all:` the API's all-fields search, were: `all:"AI scientist"`; `all:"AI scientists"`; `all:"autonomous research agent"`; `all:"autonomous scientific discovery" AND (all:evaluation OR all:benchmark OR all:failure)`; `all:"machine learning experimentation" AND (all:agent OR all:LLM)`; `all:"scientific reasoning" AND (all:agent OR all:"language model")`; `all:"AI research integrity"`; `all:"AI-generated research"`; and `all:"automated scientific discovery" AND (all:agent OR all:LLM)`. Per-query totals were 141, 143, 26, 33, 4, 221, 0, 13, and 44 records in that order (the query on research integrity returned none); the supplement stores the query log and the per-record query matches. Seven further records, found through citation chains, were added with a provenance flag, giving 453 records. Title and abstract screening, a common rescreen of 177 uncertain records, and full-text assessment of 103 prioritized papers yielded 44 primary papers. Expansion stopped at mechanism-level saturation. The map is purposive, not a probability sample of the field.

For each paper we record the unit, denominator, selection exposure, validator independence, and non-pooling boundary, giving 54 source-bounded estimands. A separate computability check asks whether each estimand supplies c, s, μ_s for the reliability target. Forty-four of the 54 instead measure a mechanism, intervention, process label, benchmark proxy, or validator property; they receive no capability bounds. Ten contribute identification components, which combine into seven target sets; two study-level funnels (FAR and MLReplicate) give two more. No new agents were run.

4.2 Cross-evaluation rankability audit

We screened all 44 papers and one recent benchmark, DeltaML-Bench [25], flagged separately: 45 paper-level records. An evaluation was eligible when at least three configurations shared a task frame and a bounded scalar endpoint, the displayed comparison was recoverable, the initiated denominator was known, and either all-unit outcomes or a sharp missing-outcome set could be reconstructed. Fifteen evaluations qualified (100 configurations, 376 unordered within-evaluation pairs); each of the 30 exclusions has a recorded paper-specific reason.

When a paper reported several endpoints, we used its explicit common scalar comparison on the widest compatible task panel. If the source contrasted a selected or best-of-k display with disclosed all-run outcomes, that contrast was used; otherwise we used the narrowest bounded endpoint recoverable for every configuration. Endpoint choices and source locations were recorded before pair statuses were computed, which prevents switching endpoints to maximize loss, but the audit was not preregistered. Several admitted endpoints are executable-task or automated-review proxies rather than scientific validity.

Each displayed strict edge is classified as certified in the same direction, reversed, or unresolved after denominator recovery. Four evaluations use selected or partially observed displays; eleven report all intended outcomes or a near-complete panel and serve as positive controls. In a complete-run evaluation the displayed and target intervals coincide, so retention follows by construction; it is an algebraic control, not independent evidence that the endpoint measures scientific validity. Certification also says nothing about the size of an order or its sampling uncertainty. The one near-complete control, Corral, lacks 3 of 1,725 intended trials for one configuration; these remain in $[0, 1]$.

5. Results

5.1 Rankability depends on observation design

Table 1 reports every admitted evaluation. Six displayed relations reverse direction and nine lose their strict order; all fifteen losses occur in selected- or missing-output designs. A reversal is not merely wider uncertainty: the denominator-aware target identifies the opposite order. ScienceAgentBench, whose headline table selects the best of three runs while an appendix discloses all-run means, supplies all six reversals and two unresolved pairs [26]; MLReplicate loses all six strict relations made by its selected display; ScientistOne has one unresolved pair [27]; BadScientist loses none [28]. These reversals concern a change of target, not a defective source: ScienceAgentBench openly evaluates best-of-three success and publishes the all-run means, and ScientistOne likewise reports both selected and average quantities. The reversals show that a best-of-three order need not carry over to the single-attempt order; they do not contradict the sources' own best-of-three results. Throughout, a displayed order is called certified, reversed, or unresolved only with respect to the all-attempt target defined here.

Table 1. Rankability audit. "Identified" counts pairs with an identified strict target order out of all pairs (the numerator and denominator of ρ); "certified" counts displayed strict orders retained in the same direction out of all displayed strict orders for the all-attempt target of this paper; a non-certified order is unsupported for that target, not an error in the source's own selected-output claim.

Evaluation	Design	Configs	Identified	Certified	Reversed	Unresolved
MLReplicate	selected	6	0/15	0/6	0	6
ScienceAgentBench	selected (best of 3)	15	103/105	95/103	6	2
ScientistOne	selected (best of 3)	5	9/10	9/10	0	1
BadScientist (rate-matched)	selected	6	15/15	15/15	0	0
DeltaML-Bench 4@6h	complete	4	6/6	6/6	0	0
MLAgentBench	complete	7	20/21	20/20	0	0
SpatialBench-Long	complete	15	90/105	90/90	0	0
SciIntegrity-Bench	complete	7	21/21	21/21	0	0
Corral task success	near-complete	6	15/15	15/15	0	0
Lab-in-the-loop (one panel)	complete	6	15/15	15/15	0	0
DiscoveryWorld	complete	3	3/3	3/3	0	0
HLER ablation	complete	4	6/6	6/6	0	0
InquiTree	complete	7	21/21	21/21	0	0
LMR-Bench	complete	6	14/15	14/14	0	0
Distributed Denial of Science	complete	3	3/3	3/3	0	0

Across the four selected-output evaluations, 119 of 134 displayed strict relations are certified, and the targets identify 127 of 145 possible pair orders; the pair-weighted gap is $G_R = 15/134 \approx 11.2\%$. The eleven controls identify 214 of 231 pair orders and certify all 214 displayed strict relations; the remaining 17 pairs were ties or overlapped at reported precision rather than lost claims. The controls span benchmark success, paper-derived task recovery, automated review,

human-gated policies, and an attack endpoint [8, 29, 11, 14, 30, 31, 32, 33, 34, 35, 25]. Pairs share systems and tasks and endpoints differ, so neither 15/134 nor the all-audit 15/348 estimates a field rate.

5.2 MLReplicate: an empty rankability graph

Six configured systems each target the same eight source papers [7]. The funnel has 48 initiated system–paper attempts, 45 generated manuscripts, 37 passing the page screen, 10 automated acceptances, and external human review (three reviewers, 1–6 scale) only for those ten; 29 of the 30 intended scores are reported, one Agent Laboratory score being missing. We normalize scores to $[0, 1]$, assign failed experiments zero, and leave every unobserved score of a completed manuscript in $[0, 1]$. The target is the mean normalized human score per initiated attempt, averaged over three intended reviews.

Table 2. MLReplicate selected means and sharp identified sets.

System	Selected mean	Identified set
CycleResearcher	0.467	$[0.117, 0.742]$
Agent Laboratory	0.257 (with missing score: $[0.240, 0.307]$)	$[0.150, 0.442]$
Tiny Scientist	0.100	$[0.025, 0.775]$
AI Researcher	0.000	$[0.000, 0.750]$
AI Scientist v1	not reviewed	$[0, 1]$
AI Scientist v2	not reviewed	$[0, 1]$

The selected means give the total order CycleResearcher > Agent Laboratory > Tiny Scientist > AI Researcher, and letting the missing Agent Laboratory score range over $[0, 1]$ does not change it. Once unreviewed manuscripts return to the denominator, the largest lower endpoint is 0.150 and the smallest upper endpoint is 0.442, so no pair satisfies $L_i > U_j$: none of the six displayed relations and none of the 15 possible orders is identified. This concerns source-configured performance under this human-score endpoint, not architecture quality at fixed compute.

5.3 Finite-design identified sets

Three independent or expert regimes point-identify an adjudicated endpoint on their realized units: 3 of 12 planned ideas in the independent AI Scientist evaluation yield an executable manuscript without an incorrect or hallucinated numerical result [10]; Aletheia yields 8/20 valid proofs under the strict rule and 9/20 under expert majority [5]; and one of three KOSMOS radiobiology hypotheses is clearly supported under an independent audit [36]. These are different endpoints on curated samples, not comparable system scores.

Other studies identify only sets. The AI Scientist completion funnel bounds independently valid compilable manuscripts per generated idea in $[0, 290/616] = [0, 47.1\%]$ [3]. In four full-pipeline attempts, one artifact reaches external review, giving an upper bound of 1/4 [37]. FAR attempts 4,717 conjectures and claims 1,050 resolutions, of which 15 author-selected artifacts are reviewed and labeled correct [6]; without calibrated validation the set is $[0, 1050/4717] = [0, 22.3\%]$, and treating the 15 labels as truth raises only the lower endpoint, to $15/4717 \approx 0.32\%$. For the similarity dimension, 18 of 50 AI-generated documents receive source-author ratings and six of these are not rated high-similarity, giving the sharp pass-rate set $[6/50, 38/50] = [12\%, 76\%]$ [12]. Assigning such intervals mechanically to all 54 estimands would be misleading: completion can bound a qualifying-artifact rate from above and selected validation can raise its lower endpoint, but a task score, process label, or uncalibrated automated judgment does neither for scientific validity.

5.4 Selection, attrition, and validation answer different questions

Selection. Luo et al. manipulate test information while holding five-candidate training and validation evidence fixed [38]. Agent Laboratory selects the validation-best candidate in 157/200 control and 87/200 manipulated decisions (a relative loss of 44.6%); AI Scientist v2 falls from 164/200 to 63/200 (61.6%). This is causal evidence about selector susceptibility in constructed tasks, not the fraction of generated findings that are false. In Aletheia's FirstProof matrix,

9 of 20 problem–agent attempts are majority-valid while the expert-preferred answer is valid for 6 of 10 problems [5]: single-attempt validity is 45% and best-of-two validity 60%, so selection accounts for $(0.60 - 0.45)/0.60 = 25\%$ of displayed success (20% if the disputed problem is counted invalid). Selection raises the chance of returning a valid proof without making the selected proof illusory. ASI-Arch's retention of 5 of 1,773 explorations for final validation (0.28%) measures search intensity and neither invalidates the five architectures nor estimates single-attempt reliability [4].

Attrition and proxies. The AI Scientist loses 326 of 616 ideas before a compilable paper; MLAgentBench fails to improve its baseline by more than 10% in 575 of 728 model–task trials; AutoResearchEval fails to exceed the published target in 178 of 240 trajectories [3, 8, 9]. These are pipeline or benchmark endpoints. A correct output can still conceal invalidity: 4 of 20 episodes in Correct Answer, Wrong Mechanism return the right output while defending a mechanism contradicted by interventions [13].

Provenance and validation. Indirect data poisoning yields an undetected poisoned conclusion in 223 of 450 runs, and a provenance audit reduces full successes to 0 of 150 [35]. Retrospectively selected false-positive configurations fail to transfer to the next model in 17 of 23 and 32 of 44 cases [39]. Grounded scrutiny finds that 22 of 88 substantive critiques arise from failed calculations, so scoring every failed run as zero broader value would be wrong [40]. These rates refer to different units and cannot be summed into a fraction of capability.

5.5 Joint identification is uncommon and coding-sensitive

Each of the 44 papers was coded on four requirements: a complete initial denominator; same-standard evaluation of all units or a probability sample; an independent validator; and an endpoint targeting scientific validity rather than a proxy. Every paper received a second coding from task files that omitted the first-round labels and rationales, with papers assigned to three partitions by a hash of the paper identifier, and all 62 criterion-level disagreements were adjudicated against the source by a coder who had not produced that paper's second code. The released records identify second coders and adjudicators only by pseudonymous labels (three of each) and do not record whether each was a person or an AI agent, their configurations or qualifications, or whether the written instruction not to consult first-round outputs was followed; blinding was by instruction and file omission and was not independently verified. The codings are therefore not independent expert assessments. Criterion-level agreement ranged from 23/44 to 32/44 (Cohen's κ from 0.217 to 0.493). The first, second, and adjudicated codings classify 3, 7, and 8 of the 44 papers respectively as satisfying all four requirements. These counts do not estimate field prevalence; disagreement concentrates at the boundaries between author-constructed tests and independent validators and between executable task success and scientific validity. A recoverable denominator alone is insufficient, and apparent joint identification depends materially on how validator independence and validity are operationalized.

In an exploratory analysis whose definitions were fixed after the adjudicated set was known, we recoded only the eight adjudicated positives. Judged against each paper's primary output claim, five endpoints reach the complete claim and three measure one dimension of it. Four designs combine claim-complete measurement with procedural (blinded, held-out, or prespecified) validator separation; none of the eight combines it with an institutionally external validator. This applies to those eight designs only and requires independent construct review.

6. Discussion

How much capability is a post-selection artifact? There is no identified global proportion. The most direct causal evidence is local: manipulated downstream information removes 44.6–61.6% of otherwise correct selections in two selectors, while transparent expert best-of-two contributes 20–25% of displayed FirstProof success without invalidating the selected proofs. Attrition, poisoning, and validator false negatives have comparable or larger magnitudes but act on different units.

Rankability certificate. An evaluation intended to estimate $\theta_{\mathcal{D},B,\pi}$ should declare the task frame, budget, policy, candidates per task, selection rule, and the information available to the selector; keep every initiated unit in the denominator; measure validity for all units or a prespecified probability sample under one reference procedure; and calibrate validator error on target-relevant positives and negatives. For a comparative claim we propose reporting a rankability certificate: the shared target $(\mathcal{D}, B, \pi, V)$ or the dimensions on which policies differ; the initiated denominator and treatment of failures; the selection and evaluation rules; validator calibration or an explicit error range; each system's identified set;

and only the directed edges with $L_i > U_j$. Incompatible targets receive no edge, and overlapping sets are reported as unresolved rather than broken by selected means. A displayed leaderboard is supported, as a statement about the shared per-attempt target, exactly when its strict order is contained in the certified partial order; a source that openly reports a different target, such as best-of-k success, is not thereby contradicted. Each certified edge holds for the realized endpoint only, and may be practically small or statistically uncertain. The supplement includes machine-readable certificates for the audit and for MLReplicate.

Limitations. The evidence map is purposive and published studies are themselves selected. The audit covers that map plus one update, not the field; its 376 pairs are dependent, its endpoints heterogeneous, and its inclusion rules and endpoint hierarchy were not preregistered. Several endpoints are proxies rather than scientific validity. Coder agreement is modest, coder identity and configuration are not documented, and the validator-separation analysis is exploratory. The identified sets are plug-in sets that omit sampling uncertainty and transport to a field-wide task distribution. One MLReplicate score is missing, although it cannot change the reported orders. All source counts were transcribed from published papers, so transcription errors would propagate; each count is linked to a source table or figure in the supplement.

7. Conclusion

Autonomous-research capability is a partially identified per-attempt quantity, not a selected-output score. Across 15 public comparative evaluations, denominator recovery preserves, reverses, or erases displayed orders; the losses concentrate in selected- or missing-output designs, with MLReplicate the limiting case of no identified order. Rankability is a property of the observation design, and a rankability certificate makes it auditable by reporting the identified partial order and leaving unsupported pairs unresolved.

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